batmonitor 0.78-py: decision tables (bm_decisions.dmn, 14 tables, 62 rows) - first matching row wins, - = any; green 1, red 0; share = input combinations the row decides (DEAD = never reached)

D7 house on AC-out1

D7								
mode_off	mode_on	over_on	under_off	under_long	on	**on**	**under**	share
1	-	-	-	-	-	**0**	**0**	32/64
-	1	-	-	-	-	**1**	**0**	16/64
-	-	1	-	-	-	**1**	**0**	8/64
-	-	-	1	1	-	**0**	**1**	2/64
-	-	-	1	-	1	**1**	**1**	1/64
-	-	-	1	-	-	**0**	**1**	1/64
-	-	-	-	-	-	**KEEP**	**0**	4/64

D7: house on AC-out1. HOUSE_ACOUT 0 = off, 1 = on, auto: on above 99 W, off after 60 s below 30 W.

D5 DO4 pulse

D5						
lockout_over	acout_over	pcc_over	grace_over	pcc_hard	**do4**	share
0	-	-	-	-	**0**	16/32
-	1	-	-	-	**1**	8/32
-	-	1	1	-	**1**	2/32
-	-	1	-	1	**1**	1/32
-	-	-	-	-	**0**	5/32

D5: DO4 pulse (FoxESS shedding): PCC > 20 kW for the grace time or > DO4_HARD_W, or grid offline and PV overfeeds one AC-out; never twice within the lockout.

D3 discharge PROP / IDLE

D3						
any_dis	house_import	prop_prev	prop_hold	deficit_gt_hold	**prop**	share
0	-	-	-	-	**0**	16/32
-	1	-	-	-	**1**	8/32
-	-	1	-	1	**1**	2/32
-	-	-	1	1	**1**	1/32
-	-	-	-	-	**0**	5/32

D3: PROPORTIONAL (1) while importing, or (already proportional / night floor) while the deficit > 20 W.

D1 Sofar first, source, winter heat pump rule

D1_WP_CAP				
winter	wp_on	b1_dis	**wp_cap**	share
1	1	1	**ZERO**	1/8
1	1	0	**MAX**	1/8
-	-	-	**NONE**	6/8

D1: winter heat pump rule. Stacks give 0 W while the Sofar Bat1 discharges, else at most WP_WINTER_W.

D1_LATCH				
b1_above_min	bat1_first	b1_ge_release	**bat1_first**	share
0	-	-	**0**	4/8
1	0	1	**1**	1/8
-	-	-	**KEEP**	3/8

D1: Sofar first latch. Off at or below BAT1_SOC_MIN, on again from BAT1_SOC_MIN + 2 %.

bat1_first

D1_B1_FIRST			
bat1_first	wp_on	**b1_first**	share
1	1	**1**	1/4
-	-	**0**	3/4

D1: Sofar first acts only while the heat pump runs.

b1_first

D1_SRC					
night	b1_first	winter	wp_on	**src**	share
0	-	-	-	**DAY**	8/16
1	1	-	-	**B1**	4/16
1	-	1	1	**WPG**	1/16
-	-	-	-	**STK**	3/16

D1: S3 source of the discharge. DAY by day, B1 = Sofar first at night, WPG = winter night with the heat pump running (no night floor), STK = night floor.

src

effect tables (bm_logic.py / bm_logic.c)

src_rule BM_SRC_RULE: the discharge per source				
src	b1_signed	trickle	idle_w	prop_hold
DAY	0	0	20	0
B1	0	0	20	0
STK	1	1	0	1
WPG	0	0	20	0

mode_rule BM_MODE_RULE: what a mode does			
mode	**action**	**capped**	**rule text**
BMS_MISSING	0 W	0	NAME
FORCE	FORCE_CHARGE_W	1	NAME(soc %)
STALE	0 W	0	NAME
LADESPERRE	0 W	0	NAME(peak h)
CHARGE	charge amount	1	NAME SRC:..
FULL	0 W	0	NAME
PROT	0 W	0	NAME(soc %)
CHARGING	0 W	0	NAME(BMS W)
DIS	discharge amount	1	(discharge path)
HOLD	0 W	0	NAME(soc %)

D2 latches and mode (one bank)

D2_LOW					
bms_ok	low_on	low_off	soc_ge_release	**low**	share
1	1	-	-	**1**	4/16
1	-	1	1	**0**	1/16
-	-	-	-	**KEEP**	11/16

D2: LOW force charge: DCL 0 or lowest cell < CELL_FORCE_V; ends once DCL is back, the cell >= CELL_FORCE_RELEASE_V and the SoC >= SOC_MIN_RELEASE.

low

D2_FORCE				
bms_ok	soc_lt_force	soc_ge_force_release	**force**	share
1	1	-	**1**	2/8
1	-	1	**0**	1/8
-	-	-	**KEEP**	5/8

D2: grid force charge, SOC_FORCE .. SOC_FORCE_RELEASE.

force

D2_PROT				
bms_ok	soc_lt_min	soc_ge_release	**prot**	share
1	1	-	**1**	2/8
1	-	1	**0**	1/8
-	-	-	**KEEP**	5/8

D2: SoC protection, SOC_MIN .. SOC_MIN_RELEASE (only while the BMS reports).

prot

D2_CHARGE_MODE				
surplus_gt_on	chg_prev	surplus_gt_hold	**charge_mode**	share
1	-	-	**1**	4/8
-	1	1	**1**	1/8
-	-	-	**0**	3/8

D2: charge mode on the surplus (> 200 W), held down to 20 W once charging.

charge_mode

D2_MODE												
bms_ok	force	low	stale	charge_mode	block	soc_lt_100	ff	prot	charging	**mode**	share
0	-	-	-	-	-	-	-	-	-	**BMS_MISSING**	512/1024
-	1	-	-	-	-	-	-	-	-	**FORCE**	256/1024
-	-	1	-	-	-	-	-	-	-	**FORCE**	128/1024
-	-	-	1	-	-	-	-	-	-	**STALE**	64/1024
-	-	-	-	1	1	-	-	-	-	**LADESPERRE**	16/1024
-	-	-	-	1	-	1	-	-	-	**CHARGE**	8/1024
-	-	-	-	1	-	-	1	-	-	**CHARGE**	4/1024
-	-	-	-	1	-	-	-	-	-	**FULL**	4/1024
-	-	-	-	-	-	-	1	-	-	**HOLD**	16/1024
-	-	-	-	-	-	-	-	1	-	**PROT**	8/1024
-	-	-	-	-	-	-	-	-	1	**CHARGING**	4/1024
-	-	-	-	-	-	-	-	-	-	**DIS**	4/1024

D2: mode of the battery, priority from top to bottom (force / low / prot = the latches above, ff = force_full_from_pv is on). 0.77-py: one bank (both stacks parallel), charged only up to the Pytes limit: no top hold window any more, SoC 100 % of the Pytes BMS = FULL.

mode

block

D4 summer peak window

D4								
enabled	season	fc_cap_ahead	surplus_cap_on	chg_prev	surplus_cap_hold	**block**	**cap**	share
0	-	-	-	-	-	**0**	**0**	32/64
-	0	-	-	-	-	**0**	**0**	16/64
-	-	0	-	-	-	**0**	**0**	8/64
-	-	-	1	-	-	**0**	**1**	4/64
-	-	-	-	1	1	**0**	**1**	1/64
-	-	-	-	-	-	**1**	**1**	3/64

D4: summer peak window, outputs block (wait below EXPORT_CAP) and cap (charge only above it).

D4_PEAK			
pcc_peak	new_day	**peak_today**	share
1	-	**1**	2/4
-	1	**0**	1/4
-	-	**KEEP**	1/4

D4: PCC above 20 kW seen today (display).

batmonitor: observed states of logic 43d60caf0fc1 (only states possible now), 07.11.2026 18:14 .. 07.11.2026 18:19 (wagodb.pv_victron, one row per minute = the state of the last 5 s cycle; generated 2026-11-07 18:19)

coverage 07.11. 18:14 .. 07.11. 18:19: 6 minutes, 6 logged with the rules (100.00 %), 0 logger off, 6 with a D1 cell (bm_d1_id since 25.10.)				
day	**logged**	**logger off**	**D1 cells**	**rule states**
Sat 07.11.	6	0	1	3
gaps (minutes without a row)				
none				

D1 cells (pv_victron.bm_d1_id) through the D1 tables of bm_decisions.dmn					
cell	**bits**	**minutes**	**set bits**	**src**	**wp_cap**
93	1+4+8+16+64	6	night, b1_above_min, b1_ge_release, winter, bat1_first	STK	none

rule states L1 | L2 | L3 (measured values in brackets removed), 3 different incl. logger off (no row / no rules)		
minutes	**%**	**L1 | L2 | L3**
4	66.7	PROP SRC:STK EQ | PROP SRC:STK EQ | PROP SRC:STK EQ
1	16.7	CHARGE SRC:STK EQ | CHARGE SRC:STK EQ | CHARGE SRC:STK EQ
1	16.7	PROP SRC:STK 1PH | PROP SRC:STK STBY | PROP SRC:STK STBY